INDUSTRIAL PROJECT REPORT

HR Data Analytics

Industry Project Title	Employee Attrition Analysis
Name of the Company	TCS iON
Name of the Institute	Guru Nanak College of Arts, Science & Commerce

Start Date	End Date	Total Effort (hrs.)	Project Environment	Tools used
16 April 2026	21 April 2026	40 hrs	Windows	Python (pandas, seaborn,scikit-learn), Power BI, Excel

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1. Acknowledgements

I sincerely thank my faculty mentor and institute for their continuous guidance and encouragement throughout this project. Their support motivated me to push beyond my limits and successfully complete both the Data Analytics course and the internship project

2. Objective and Scope

The objective of this project is to analyze employee attrition patterns and identify key drivers such as salary, overtime, job satisfaction, and work-life balance.

Scope includes:

- Cleaning and preparing HR dataset (1480 records, 38 columns).
- Building interactive dashboards in Power BI.
- Applying machine learning models (Logistic Regression, Decision Tree, Random Forest).
- Providing actionable insights for HR decision-making.

3. Problem Statement

- HR departments often struggle with scattered data across multiple spreadsheets, making attrition analysis slow and inefficient. This project addresses the problem by centralizing HR data, cleaning inconsistencies, and building dashboards + predictive models to provide clear, data-driven insights.

4. Existing Approaches

Before implementing Power BI dashboards and ML models, HR teams typically relied on:

1. **Excel Spreadsheets**
   - HR maintained employee records in multiple Excel files.
   - Reports were static, prone to errors, and lacked interactivity.
   - Difficult to handle large datasets efficiently.
2. **HRIS Systems (SAP, Workday, BambooHR)**
   - Provided basic built-in reports.
   - Customization required IT support.
   - Visualizations were rigid and not suitable for strategic decision-making.
3. **Static PDF/PowerPoint Reports**
   - Monthly reports prepared manually.
   - Outdated the moment they were published.
   - No ability to filter or drill-down into data.
   -

📘 Why Power BI Was Chosen

- Fully interactive visuals with filters and drill-down.
- Direct connection to CSV and other data sources.
- Custom DAX measures for advanced calculations.
- High-quality visuals suitable for all stakeholders.
- Cost-effective (free desktop version).

5. Approach / Methodology - Tools and Technologies used

- **Tools Used:** Python (pandas, seaborn, scikit-learn), Power BI.

- **Steps:**

- Import libraries.
- Load dataset (1480 × 38).
- Handle missing values (YearsWithCurrManager → median).
- Drop redundant columns (EmployeeNumber, EmployeeCount, StandardHours, Over18).
- Encode categorical variables.
- Build dashboards in Power BI.
- Train ML models (Logistic Regression, Decision Tree, Random Forest).

6. Workflow

- **Data Loading:** Imported HR_Analytics.csv into Python and Power BI.

- **Data Cleaning:** Null values handled, categorical encoding applied.

- **Visualization:** Attrition distribution, Overtime vs Attrition, Salary vs Attrition, JobRole vs Attrition.

- **Dashboard:** Age, JobRole, Department, Gender, Overtime, WorkLifeBalance, Salary, JobSatisfaction.

- **Modelling:** Logistic Regression (Accuracy ~0.85), Random Forest (Accuracy ~0.86, ROC-AUC ~0.88).

- **Feature Importance:** MonthlyIncome, DistanceFromHome, OverTime, YearsAtCompany identified as top drivers.

Assumptions

- The dataset provided (HR_Analytics.csv) is **accurate and complete**, representing real employee records.

- All missing values can be reasonably imputed using statistical methods (median for numeric fields).

- Attrition is treated as a **binary variable** (Yes = 1, No = 0).

- Employees' responses to satisfaction and work-life balance ratings are assumed to be **honest and unbiased**.

- Salary slabs and monthly income values are assumed to be **consistent across departments**.

- The machine learning models (Logistic Regression, Decision Tree, Random Forest) are assumed to be **sufficient for predictive analysis** without requiring deep learning.

- Power BI dashboards are assumed to be viewed by HR managers with **basic data literacy**

7. Implementation - Data collection, Processing Steps, Diagrams - Charts, Table

- **Attrition Distribution:** Majority employees stayed, ~200 left.

- **Overtime:** Employees with overtime showed higher attrition.

- **Salary:** Lower salary groups had higher attrition.

- **Job Role:** Sales Representatives and Laboratory Technicians showed higher turnover.

- **Work-Life Balance:** Poor balance strongly linked to attrition.

- **ML Models:** Random Forest performed best with balanced accuracy and feature importance analysis.

8. Solution Design

- **Page 1:** Workforce Overview (KPIs, Age, Gender, Department).

- **Page 2:** Performance & Compensation (Salary benchmarking, Job Satisfaction, Engagement).

- **Page 3:** Attrition & Diversity (Attrition by Department, Overtime, Work-Life Balance, Demographics).
- **Page 4:** Key Insights & Recommendations (text boxes summarizing findings). This design ensures HR managers can navigate easily and focus on specific workforce dimensions.

9. Challenges & Opportunities

Challenges:

- Handling missing values in YearsWithCurrManager.
- Encoding categorical variables for ML models.
- Attrition imbalance (majority employees stayed, fewer left).

Opportunities:

- Connect dashboard to live HRIS database for real-time updates.
- Add predictive attrition scoring with ML.
- Role-based dashboards for managers.

10. Reflections on the project

- This project gave hands-on experience in data cleaning, visualization, and machine learning. It highlighted the importance of preprocessing (null handling, encoding) and the impact of clear dashboards on decision-making. Iterating on chart types and ML models improved my understanding of HR analytics and predictive modelling.

11. Recommendations

- **Attrition:** Sales & Lab Technicians show higher attrition → conduct stay interviews, review workload.
- **Compensation:** Lower salary groups → consider salary revision.
- **Engagement:** Poor work-life balance and overtime → introduce flexible policies.
- **Performance:** Employees on PIP → structured coaching and mentoring.
- **Diversity:** Increase investment in diverse hiring channels.

12. Outcome / Conclusion

- The project successfully delivered a Power BI dashboard and predictive models that highlight attrition drivers. Insights show that salary, overtime, job satisfaction, and work-life balance are critical factors influencing employee retention. This project

strengthened my technical skills in data analytics and visualization, and provided valuable industry exposure.

13. Enhancement Scope

- Real-time HRIS integration.
- Predictive attrition scoring with advanced ML.
- Role-based dashboards for managers.
- Time-series analysis of attrition trends.

14. Link to code and executable file :

15. Research questions and responses

- **Which department has highest attrition?** → Sales & Lab Technicians.
- **Does overtime affect attrition?** → Yes, overtime employees show higher attrition.
- **Is salary linked to attrition?** → Lower salary groups have higher attrition.
- **Which features are most important?** → MonthlyIncome, DistanceFromHome, OverTime, YearsAtCompany.

16. References

- HR_Analytics.csv dataset (provided).
- Microsoft Power BI Documentation.
- Scikit-learn Documentation.
- DAX Reference Guide.
- SHRM HR Metrics Handbook.